

Chalk (Calcium, Carbon and Oxygen)



Sugar (Carbon, Hydrogen and Oxygen)

3.4.1 Properties of Compounds

- A compound is formed only when the constituent elements combine in a fixed proportion.
- The properties of a compound are different from those of its constituent elements.
- A compound cannot be broken down by physical methods. This is because a compound is made up of different elements that are chemically combined. Sodium chloride cannot be separated by physical methods such as filtration.
- A compound can be separated into its constituent elements by chemical methods only.

ACTIVITY 2

Complete the following table.

Compound	Constituent Elements
Water	
Salt (Sodium chloride)	
Sodium carbonate	
Baking soda (sodium bicarbonate)	
Sugar	
Calcium oxide	
Calcium hydroxide	
Sodium hydroxide	
Potassium hydroxide	

ACTIVITY 3

Complete the following table.

Formula	No. of different elements	Name of Elements
H_2O	H – 2 O – 1	Hydrogen, Oxygen
NaCl		
$C_6H_{12}O_6$		
NaOH		

Table 3.4 Difference between an element and a compound

Elements	Compounds
An element is the simplest substance.	A compound is a chemical substance formed by the combination of two or more elements.
Elements combine to form compounds.	Compounds can be split into elements.
Atoms are the fundamental particles of an element.	Molecules are the fundamental particles of a compound.

3.5 Chemical Formulae

Often we write water as H_2O. This is the chemical formula for water molecule. This means that each molecule of water has two hydrogen atoms combined with one oxygen atom. A chemical formula is a symbolic representation of one molecule of an element or a compound. It provides information about the elements present in the molecule and the number of atoms of each element. In H_2O, small number beside the 'H' is called subscript. It tells us the number of atoms of that element present in the molecule. Hence, there are two hydrogen atoms in water molecule. There is no number of besides 'O'. It means that there is only one atom of that element present in the molecule. Hence, there is 1 oxygen atom in a water molecule. Can you guess the types of atoms and number of each of the atoms in sodium chloride? Which is the chemical formula for cooking salt?